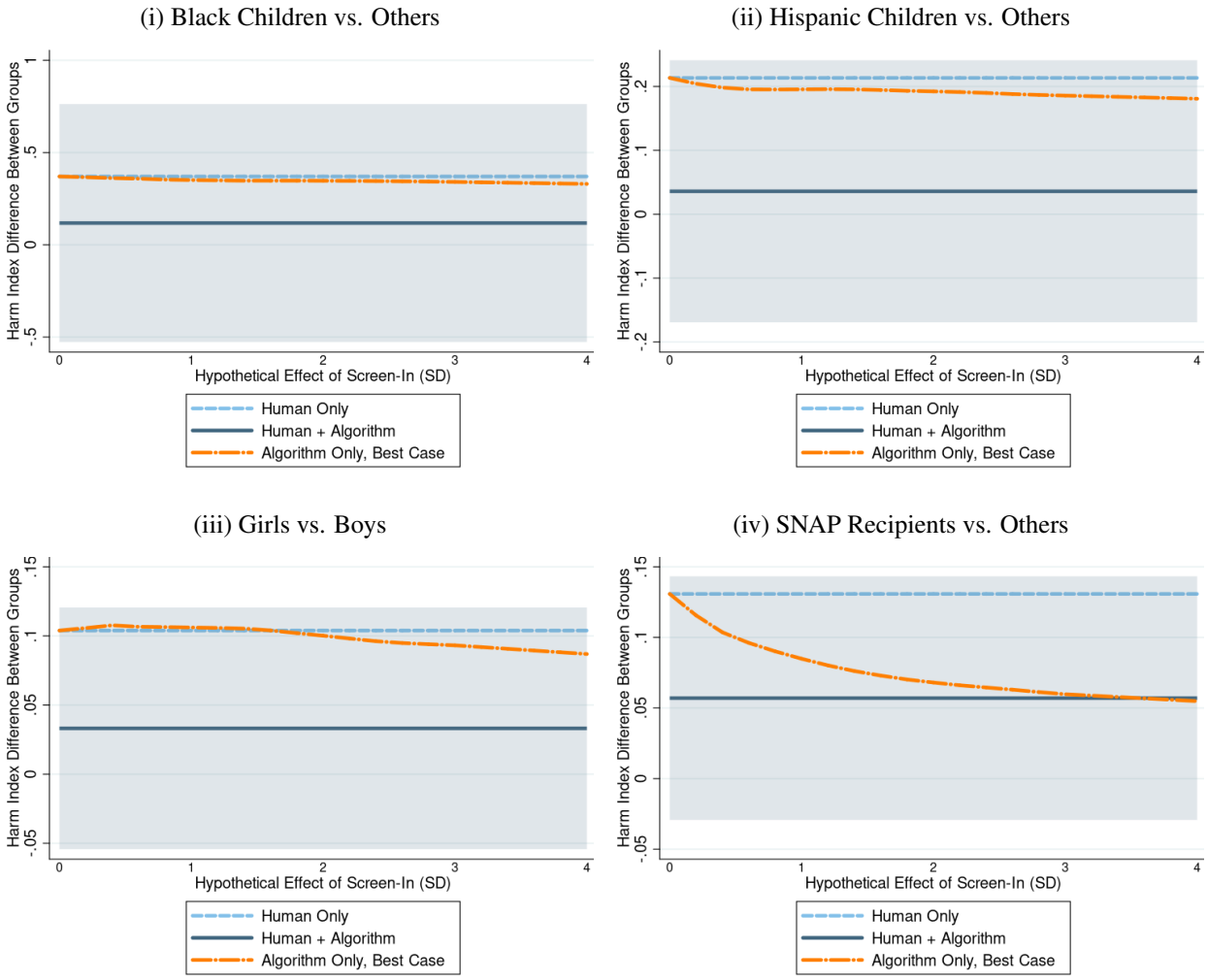
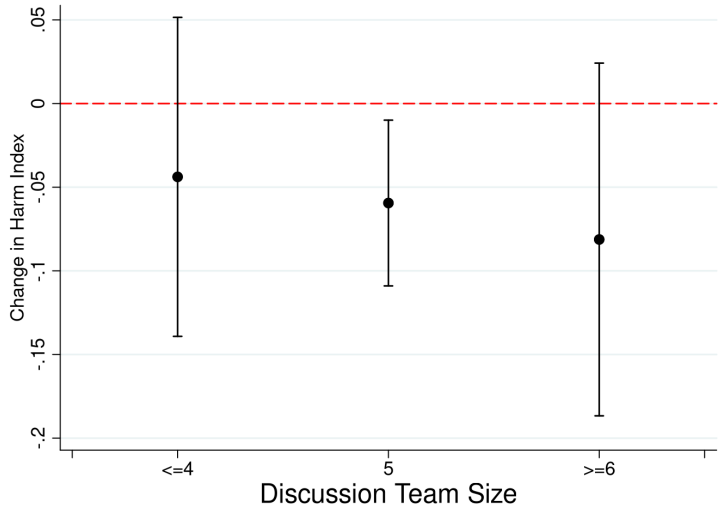
Figure A.18: Health Disparities by Child Subgroup with Algorithm-Only Counterfactual



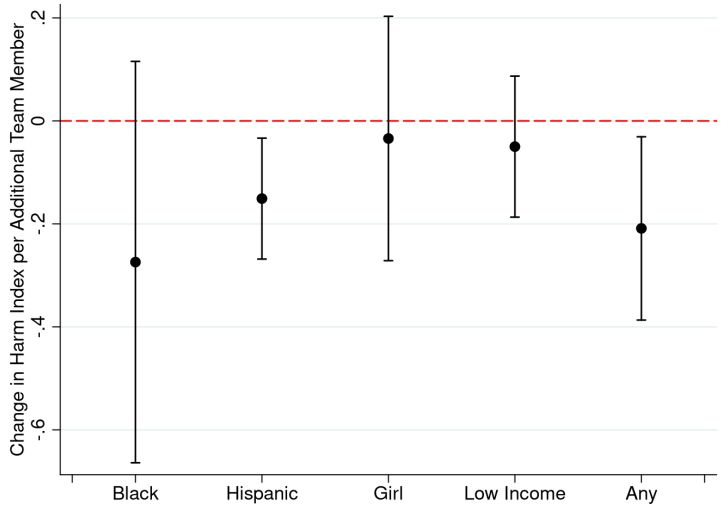
Notes: For each child subgroup as reported in the title of each figure, we plot the health disparities in child harm for three different scenarios: when humans do not have access to the algorithm (light blue short-dashed line) which is our control group, when humans have access to the algorithm (solid dark blue line, with shaded area representing a 90% confidence interval; main estimates presented in Table 1), and the best-case scenario for the algorithm (orange dash-dot line). We assume that the effects of a screen-in are homogeneous across groups. The best-case scenario for the algorithm assumes that the children who were screened in by humans, but for whom the algorithm would have screened out, would have been just as well off under the algorithm decision. Effects of the algorithm screening in children who humans screened out depends on the hypothetical benefits of a screen-in on the child harm index (horizontal axis). SNAP (food stamp) eligibility is a proxy for lower socioeconomic status.

Figure A.19: Impacts of Algorithm on Harm Index, by Team Size



Notes: This figure shows estimates of the impact of access to the algorithm tool on the standardized index of child harm (vertical axis), by size of the discussion team making the initial screen-in decision (horizontal axis). Randomization procedure controls are included throughout, as well as unit and month-year fixed effects. Standard errors are clustered at the household level and are shown at the 90 percent confidence level.

Figure A.20: Impact of Larger Team on Algorithm's Harm Reduction for Disadvantaged Groups



Notes: This figure shows a disproportionate improvement in algorithm treatment effects for members of disadvantaged groups with larger teams. The standardized index of child harm is regressed on a triple-interaction of algorithm availability (binary), demographic group (binary), and team size (continuous). The vertical axis indicates the coefficient estimate and standard error bounds on the triple-interaction term, and the horizontal axis indicates the demographic group that is compared to all children not in that group. "Any" consists of all children in the sample who fall into at least one of the disadvantaged categories. Randomization procedure controls are included throughout, as well as unit and month-year fixed effects. Standard errors are clustered at the household level and are shown at the 90 percent confidence level.

Table A.1: Sample Restrictions for Trial Analysis Sample, November 2020 – March 2022

Restrictions	**Observations**	**Unique Children**	**Referrals**
1. Referral During Trial Period	15,023	7,723	8,309
2. Scored	13,115	7,021	7,345
3. First Scored Referral was in Trial	6,852	6,852	4,213
4. Excluding Children Seen in Testing Phase (Phase 1) of Trial	6,601	6,601	4,073
5. Excluding Children with Household Member already in Trial	5,998	5,998	3,631
6. First Call going to Discussion Team	5,162	5,162	3,067
7. Omit if Multiple Scored Referrals before First Discussion Team Meeting	4,919	4,919	2,935
8. Final Sample, Excluding Observations with Overwritten Status	4,681	4,681	2,832

Notes: This table presents the sample restrictions used to arrive at the main analysis sample. The trial ran from November 1, 2020 until March 29, 2022. Over this time period, the county received $8,309$ referrals involving $15,023$ child-referral observations. The primary unit of analysis is at the child level: children are each randomized once, during their first referral in the trial. $6,852$ unique children were randomized and successfully scored.[20] Approximately a third of randomized children are omitted from the analysis sample for various reasons. First, we exclude $N = 251$ children who were first scored during a pilot phase prior to the rollout of the full trial, as teams may have remembered prior scores for the subset of these children assigned to the control group. Children with a sibling already randomized during the trial were also excluded from the analysis, as a sibling's prior assignment could have already impacted the newly randomized child prior to their own assignment ($N = 603$). We restrict our sample to standard calls assigned to the normal team meeting procedure, and therefore omit $N = 836$ non-standard calls that were not seen by a normal discussion team, such as calls listing concerns about a foster family or calls sent to an overflow discussion team where the tool was not available. Given the complications with implementing randomization discussed in Appendix C, we make two further sample restrictions to guarantee random assignment. First, we omit a small subset of referrals where there were multiple calls placed for a family before a team meeting (e.g., a family referred on Friday evening, re-referred on Saturday, and both referrals seen by a team on Monday). In practice, some of these children were re-randomized with multiple treatment statuses before first being discussed, and we therefore omit the entire group ($N = 243$). Second, we exclude all children whose original status was overwritten as a result of the implementation issue from Nov 2020-Mar 2021. We thus omit an additional $N = 238$ children. Our remaining analysis sample consists of $4,681$ unique children from $2,832$ referrals.

[20] Nine percent of calls could not be scored due to being made during evening (non-standard) hours or being assessed as high emergency by a hotline operator. In order to generate a score for each child, the child's referral needs to be transferred to the state central data system. This happens automatically at the end of each work day: the process begins around 5pm local time and typically takes 12 to 14 hours to complete. Emergency referrals that are immediately screened in will therefore not have had time to be copied over and have a score generated. Furthermore, referrals that arrive approximately after 5pm and before 8am the following morning will be reviewed by a team before information is copied into the main data system, and therefore will not have a score generated. On certain days, the main data system was down or the transfer process took longer than usual and the team started reviewing referrals before a score could be generated. Therefore scores are not generated for every referral.